

KHEYDZHAN, Russian FederationWMO: **310830**Lat: **59.967N**Lon: **144.783E**Elev: **397**StdP: **14.49**Time Zone: **10.00 (E10)**Period: **86-15**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.1	-44.4	-51.2	0.3	-47.7	-47.8	0.3	-44.3	9.3	6.5	7.8	4.6	2.0	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	78.8	61.9	75.1	60.4	71.4	59.0	63.7	74.8	61.8	72.4	60.0	69.1	5.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	76.8	66.9	57.5	71.5	63.8	56.1	67.9	62.4	29.3	74.6	27.9	73.0	26.7	69.4	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.4	7.9	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.5	-16.2	-8.4	3.9	23.5	38.5	52.6	57.5	55.2	42.5	21.5	-1.4	-12.8
	DBStd	28.88	16.07	14.00	11.89	8.13	5.71	5.82	4.47	5.62	7.45	12.37	16.01	17.38
	HDD50	10948	2052	1635	1430	795	356	40	1	20	244	884	1543	1948
	HDD65	15878	2517	2055	1895	1245	820	373	237	306	675	1349	1993	2413
	CDD50	549	0	0	0	0	1	117	232	180	19	0	0	0
	CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0
	CDH74	407	0	0	0	0	0	143	194	70	0	0	0	0
	CDH80	59	0	0	0	0	0	20	36	3	0	0	0	0
Wind	WSAvg	2.6	1.9	2.1	2.5	3.0	3.8	3.8	3.0	2.5	2.2	2.1	2.3	2.2
Precipitation	PrecAvg	27.7	1.0	0.5	0.9	1.4	1.8	2.4	3.6	4.8	4.1	3.5	2.6	1.2
	PrecMax	40.6	3.1	2.0	3.3	3.8	5.0	4.3	8.9	11.6	9.0	6.7	11.1	6.4
	PrecMin	13.7	0.0	0.1	0.1	0.1	0.3	0.5	0.8	1.8	1.5	0.5	0.5	0.0
	PrecStd	6.3	0.8	0.4	0.8	0.9	1.0	1.1	1.8	2.5	2.2	1.9	2.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	28.4	34.7	48.2	66.3	83.6	83.8	79.7	69.7	50.0	33.3	25.0
		MCWB	22.9	25.9	28.2	37.7	50.6	61.9	67.3	62.3	56.8	39.9	32.0	23.4
	2%	DB	18.3	24.6	30.2	43.7	61.6	78.2	79.2	76.0	65.1	44.4	29.2	22.4
		MCWB	16.3	22.8	25.5	34.2	47.4	59.8	63.6	61.7	53.5	38.3	28.2	21.0
	5%	DB	13.4	19.5	27.3	40.7	57.8	74.3	74.9	72.7	61.3	40.6	26.5	18.8
		MCWB	11.8	17.3	22.6	32.6	45.9	57.5	62.1	60.3	52.4	36.1	25.0	16.8
	10%	DB	9.5	13.9	24.4	37.5	53.6	70.2	70.5	68.8	57.3	37.8	23.6	13.9
		MCWB	8.1	11.5	20.7	30.9	43.4	55.9	60.2	58.2	50.2	34.7	22.3	11.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	26.9	29.2	38.6	51.8	64.1	69.5	65.5	58.5	41.9	33.2	23.8
		MCDB	23.8	27.9	33.6	46.2	64.0	76.1	82.1	75.3	67.2	46.6	34.3	24.3
	2%	WB	16.1	23.1	25.4	34.6	48.6	61.1	64.9	62.9	56.4	39.5	28.4	21.2
		MCDB	18.0	24.8	29.9	41.1	59.8	75.6	75.9	72.7	62.1	42.5	29.1	22.4
	5%	WB	12.0	17.3	23.1	33.2	46.0	58.9	62.5	61.1	53.6	37.2	25.3	16.6
		MCDB	13.6	19.7	26.8	38.6	56.8	72.0	73.5	70.2	59.4	40.1	26.1	18.4
	10%	WB	8.1	11.5	20.8	31.8	43.7	56.6	60.7	59.3	50.9	34.8	22.4	11.4
		MCDB	9.6	13.9	24.5	36.7	52.7	68.8	69.5	66.8	56.5	37.3	23.8	13.5
Mean Daily Temperature Range	5% DB	MDBR	19.7	28.8	30.3	23.9	21.6	26.7	18.9	22.7	22.6	20.9	20.4	18.4
		MCDBR	15.9	20.8	31.2	27.9	32.4	40.1	34.3	33.5	30.2	21.0	12.1	14.0
		MCWBR	14.7	19.0	25.4	19.0	19.1	22.0	19.5	19.8	18.8	14.6	11.7	13.5
	5% WB	MCDBR	15.8	17.5	26.7	22.4	30.8	36.1	30.1	27.3	25.3	17.6	12.5	14.6
		MCWBR	14.8	16.1	22.2	15.2	18.4	20.2	18.0	16.8	16.3	12.9	12.2	13.9
Clear-Sky Solar Irradiance	taub		0.229	0.273	0.334	0.454	0.467	0.430	0.483	0.431	0.347	0.311	0.250	0.209
	taud		2.040	2.071	1.951	1.692	1.852	2.202	2.069	2.212	2.413	2.192	2.035	1.955
	Ebn at Noon		173	226	242	227	238	254	234	239	246	208	155	128
	Edh at Noon		14	25	40	62	57	41	46	37	26	22	14	10
All-Sky Solar Radiation	RadAvg		147	463	1028	1598	1689	1676	1466	1189	849	490	200	76
	RadStd		12	27	42	123	157	153	223	137	115	54	23	6